Strong Tree-Childness Is a Sharp Generic-Identifiability Boundary for Level-2 Jukes–Cantor Networks

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Abstract

Reticulate evolutionary histories encode hybridization, recombination, and lateral transfer, but such histories are scientifically recoverable from sequence data only when distinct networks induce distinguishable site-pattern distributions. We determine this information limit for binary, LSA-rootable, already-simple semi-directed strongly tree-child level-2 networks under the fixed one-step reticulation-preserving semi-directed convention used here and the open four-state Jukes–Cantor model. A source-relative full-dimensional regular stochastic containment occurs if and only if the labelled reduced bridge trees agree and every pair of corresponding blobs is labelled-isomorphic or differs by ordinary triangle redirection. Hence no proper one-sided generic containment exists, and an exact generic distribution determines a unique topology modulo that redirection. Fourier cut ranks recover the bridge tree, while positive rank-one factorizations recover only projective local tensors modulo the full incidence-scaling group; physical bridge multipliers are not identified. A graph-to-algebra classification of cycle and theta factors, together with marginal submersions and coherent bounded probes, reconstructs arbitrary subdivisions.

The boundary is sharp even without triangles. For every $n \geq 4$ we construct two triangle-free networks that are weakly but not strongly tree-child, nonisomorphic, and not triangle-related, yet whose open JC images share a regular relatively open subset of full dimension $2n + 1$. A second triangle-containing family exhibits a distinct pendant-transfer mechanism. All finite-atlas, class-membership, Fourier-equality, sign, and Jacobian claims are supported by exact primary and separately implemented replay certificates.

Keywords. phylogenetic networks; identifiability; Jukes–Cantor model; level-2 networks; strongly tree-child networks; algebraic statistics.

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1 Introduction

Phylogenetic-network identifiability asks which evolutionary histories are recoverable from an exact site-pattern distribution. Even when the substitution process on every edge is simple, reticulation choices make a network model a structured mixture over displayed trees. The central issue is therefore not merely whether one polynomial ideal differs from another, but whether two *open stochastic* model images share a region of full dimension.

For level-2 networks, three phenomena set the boundary of the problem. First, ordinary redirection of a triangle can preserve a full-dimensional local JC germ. It is therefore necessary to quotient by that move. Second, we construct a four-leaf theta pair with a more serious pendant-transfer ambiguity in the weakly tree-child class. That pair is not strongly tree-child, so it leaves open the natural sharp question: does strong tree-childness remove every generic ambiguity other than triangle redirection? Third, the exact Omega pair audited here shows that weak-class failure does not require a triangle at all.

We answer yes under the fixed-mixed-graph, reticulation-preserving semi-directed convention stated below. Our result strengthens a triangle-free level-2 theorem of Englander et al. [6] by allowing triangles and identifying their exact quotient. It also refines generic distinguishability by classifying the directed full-dimensional containment relation and by giving a pointwise cut criterion and constructive reconstruction. The result parallels the role of triangle quotients in level-1 identifiability [3], while the proof must handle the two-reticulation theta core directly. Related local and semialgebraic phenomena appear in work of Ardiyansyah [1], Cox, Gross, and Martin [4], Currie et al. [5], and Sullivant [15]. Invariant-based network separation was developed for large-cycle JC models by Gross and Long [8], and Hollering and Sullivant [9] introduced an algebraic-matroid/Jacobian strategy for generic phylogenetic identifiability. Complementary combinatorial work reconstructs outer-labeled-planar, galled level-2 networks up to a canonical form from displayed quartets and inter-taxon quartet distances [10]. The present result instead concerns full-dimensional containment of open JC model images, does not assume galledness, and recovers the standard semi-directed topology modulo ordinary triangle redirection.

Precise comparison with Englander et al. In version 4, their Theorem 3.2 proves generic JC distinguishability for binary, triangle-free, strongly tree-child level-2 semi-directed networks, where distinguishability means that the exceptional common-parameter sets have measure zero in each model. Their Theorem 3.1(a) gives the direct level-1 parallel: identifiability up to the reticulation choice in a triangle. Here triangle-freeness is removed at level two and ordinary T is shown to be the exact remaining quotient. We additionally classify the directed full-dimensional containment relation, prove a pointwise open-domain cut criterion, and obtain structural reconstruction. The Omega family shows that the strong hypothesis is sharp even within the triangle-free weak class. These are refinements of their distinguishability statement, not a claim that the two definitions are logically identical.

The proof has four layers. A pointwise Fourier flattening theorem recovers all bridge splits even under one-sided containment. Rank-one cut factorizations then recover each local boundary tensor only projectively; the correct fiber is the full incidence-scaling action, not one reciprocal scalar per bridge. A structural generator theorem reduces every level-2 blob to a cycle or one of four directed theta cores. Finally, exact graph-derived invariants classify a rigid bounded support of every core, and a submersion plus overlapping one- and two-port probes reconstructs arbitrary subdivision words. This finite step is computer-assisted, but its theorem object is a decorated directed relation carrying both graphs, all port matches, all incoming roles, and a complete graph-to-polynomial provenance chain.

Our three principal statements are as follows.

Theorem 1.1 (Strong-class classification)**.** *Let N, N' be leaf-labelled binary standard semi-directed strongly tree-child level-2 networks on the same taxon set. Under the open Jukes–Cantor model,*

$$N \preceq_{\mathrm{JC}} N' \iff \text{their labelled reduced bridge trees agree, and every pair of corresponding nontrivial blobs is labelled-isomorphic or differs by ordinary triangle redirection } \mathsf{T}.$$

Consequently there is no proper one-sided generic containment. Moreover, $N \bowtie_{\mathrm{JC}} N'$ holds exactly under the same condition.

Corollary 1.2 (Generic identifiability modulo triangles)**.** *For every fixed N in the class there is a proper algebraic subset $E_N \subsetneq \mathcal{V}_N^{\mathrm{JC}}$ such that every distribution $p \in \mathcal{M}_N^{\mathrm{JC}} \setminus E_N$ determines the labelled standard semi-directed topology of N modulo ordinary triangle redirection.*

Theorem 1.3 (Triangle-free sharpness)**.** *For every $n \geq 4$ there are two leaf-labelled binary standard semi-directed* triangle-free *level-2 networks $\Omega_n, \Omega'_n \in W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$ such that:*

(i) they are nonisomorphic and not related by ordinary triangle redirection;
(ii) each open JC model image has dimension $2n+1$; and
(iii) their images share a regular relatively open subset of dimension $2n+1$.

Thus strong tree-childness is a sharp boundary against the full weakly tree-child class, independently of triangle ambiguity. A second all-n family below retains the triangle-containing pendant-transfer mechanism and has common dimension $2n$. The qualifier "generic" in corollary 1.2 means the complement of a proper algebraic exceptional set. The local triangle statement is different: we prove one common full-dimensional regular Euclidean-open germ, not equality or density of the complete stochastic images.

Both all-n sharpness families are obtained from certified four-leaf base collisions by the positive analytically invertible identical-cherry substitution of section 9.

2 Conventions and observational relations

2.1 Rooted and semi-directed networks

Definition 2.1 (Fixed-mixed-graph semi-directed convention)**.** *Let $\mathcal{X}$ be a finite set of taxon labels. An* LSA-valid rooted binary phylogenetic network *on $\mathcal{X}$ is a finite directed acyclic graph without parallel arcs in which the root has bidegree $(0,2)$, a tree vertex has bidegree $(1,2)$, a reticulation has bidegree $(2,1)$, and each leaf has bidegree $(1,0)$ and a unique label in $\mathcal{X}$. Every vertex is reachable from the root. The root is the lowest stable ancestor: it lies on every root-to-leaf path and no proper descendant does so for every labelled leaf. A rooted presentation is* tree-child *if every internal vertex has a tree vertex or leaf as a child.*

We use one reticulation-preserving semi-deorientation, denoted sd_0. Mark every arc entering a reticulation, undirect every other arc, delete the binary root, and replace its two incident edges by one edge between its children, retaining at either endpoint exactly the arrowhead present before deletion. A presentation is admissible only if this single suppression produces a simple binary mixed graph: no loop or parallel edge is created and every reticulation and arrowhead survives. No later degree-two or parallel cleanup is part of sd_0. Isomorphisms preserve labels, undirected edges, retained arrowheads, and vertex roles. An admissible rooting *of a mixed graph is an LSA-valid rooted binary network whose sd_0 image is exactly that graph. For the already-simple mixed graphs compared here, this agrees with the rooted-partner convention of Englander et al. [6, Definitions 2.1–2.3] and is the binary LSA-valid specialization of Holtgrefe et al. [11]; it is not a broader exhaustive cleanup after taking an induced subnetwork.*

A simple binary mixed graph admitted by definition 2.1 will be called a *standard semi-directed network* throughout this paper. Here and below "standard" refers only to this fixed-mixed-graph

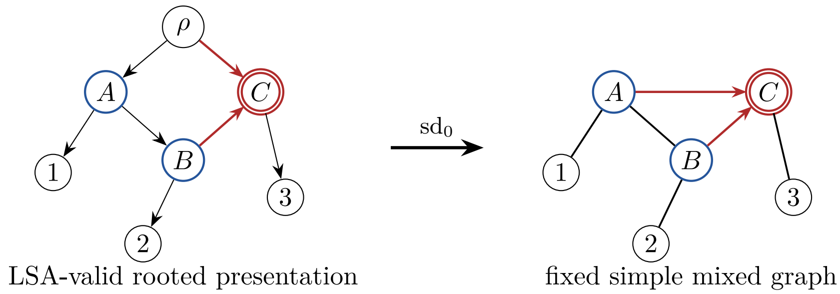


Figure 1: The fixed-mixed-graph convention. Only arcs entering a reticulation retain arrowheads. The binary root is deleted and its two incident edges are replaced once; the result must already be simple and binary. No iterative cleanup or hidden rooted refinement is admitted.

convention; it does not assert that every semi-directed convention in the literature is literally identical.

Four operations must therefore be kept distinct. Formation of the full semi-directed topology uses only the one root suppression above. Restriction to a leaf subset and construction of a displayed network may exhaustively delete unlabeled dead ends and suppress degree-two vertices. Neither operation inserts a new rooted refinement. In particular, exhaustive cleanup used after restriction does not enlarge the admissible-rooting set of one fixed mixed graph.

We distinguish

$$\begin{aligned} R_{\mathrm{TC}} &= \{\text{chosen rooted tree-child presentations}\}, \\ W_{\mathrm{TC}} &= \{\text{mixed graphs with at least one tree-child admissible rooting}\}, \\ S_{\mathrm{TC}} &= \{\text{mixed graphs with at least one admissible rooting, all of which are tree-child}\}. \end{aligned}$$

Lemma 2.2 (No-omnian criterion)**.** *Let N be a standard semi-directed network with an admissible rooting. Then $N \in S_{\mathrm{TC}}$ if and only if every tail of a retained reticulation edge is incident with two undirected edges.*

This is exactly the incidence criterion stated for strongly tree-child semi-directed networks by Englander et al. [6, Definition 2.3]; the proof below records its binary, LSA-valid, fixed-sd_0 specialization.

Proof. Suppose first that a tail v fails the incidence condition. Binary degree leaves two possibilities. Either v is itself reticulate and has a reticulation child, or v is an ordinary vertex tailing two retained edges. In the first case v has a reticulation child in every admissible rooting. Indeed, inserting the root on that retained edge would give the root two reticulation children and would suppress to an edge with arrowheads at both ends, not to the fixed one-arrowhead edge. Thus every admissible rooting is already non-tree-child. In the second case, any rooting not subdividing one of the two retained edges makes v a tree vertex with two reticulation children. Suppose instead that an admissible tree-child rooting subdivides one of them. Its other root child is v, whose remaining children are the other reticulation and the endpoint u of its unique undirected incidence. Move the root from the retained edge to the ordinary edge vu. The old root suppresses to the first retained edge, all other arrowheads remain fixed, and no directed cycle is created: in the original rooting v had indegree one from the root, so no descendant of either reticulation could reach v. The new root is the LSA. To see this directly, follow tree or leaf children from u to a leaf; that all-tree path has no second entrance, while a path through either reticulation child of v avoids every vertex of it. Hence no proper descendant of the new root lies on every root-to-leaf path. In the new admissible rooting both children of v are reticulate. Thus some admissible rooting is not tree-child, and $N \notin S_{\mathrm{TC}}$.

Conversely, assume the incidence condition. In any admissible rooting, a nonroot tree vertex with no reticulation child plainly has a tree or leaf child. A tree vertex with one reticulation child has, besides that retained edge, two undirected incidences; one is its parent and the other is a tree or leaf child. A reticulation cannot tail a retained edge, because its two incoming retained edges and that outgoing retained edge would leave it no undirected incidence. It therefore has a tree or leaf child. If the root is inserted on an ordinary edge, at least one endpoint is a tree vertex or leaf. If it is inserted on an edge entering a reticulation, the tail endpoint has two undirected incidences and hence is an ordinary tree vertex, giving the root a tree child. Finally, a root with two reticulation children would suppress to an edge with two arrowheads, which is not admitted by definition 2.1. Every admissible rooting is therefore tree-child. This is the binary fixed-graph specialization of Holtgrefe et al. [11, Theorem 5]. □

A cut edge is an edge whose deletion disconnects the underlying graph. A *blob* is a maximal nontrivial biconnected block of the underlying simple graph. Equivalently in the present binary setting, the nontrivial blobs are the non-bridge blocks in the bridge decomposition. A network is *level two* when every blob contains at most two reticulations. Contracting each nontrivial blob and retaining ordinary trivalent components gives the reduced, leaf-labelled *tree-of-blobs*, called the *bridge tree* below.

Definition 2.3 (Ordinary triangle redirection)**.** *An ordinary triangle redirection* T *changes only which vertex of a triangle is reticulate and the two arrowheads entering it. It leaves the labelled underlying graph, every pendant-component placement, and every arrowhead outside the triangle unchanged.*

The generated local relation is denoted $\sim_{\mathsf{T}}$. Both endpoint topologies in any invocation below are required to belong to the theorem's class.

2.2 The open JC model

Identify the four states with $\mathcal{G} = \mathbb{Z}_2 \times \mathbb{Z}_2$. Every edge e has a single nontrivial Fourier multiplier $x_e \in (0, 1)$, so its transition matrix is

$$T_e(a,b) = \begin{cases} (1+3x_e)/4, & a = b, \\ (1-x_e)/4, & a \neq b. \end{cases}$$

Every reticulation r has an inheritance probability $\lambda_r \in (0, 1)$, and the root state is uniform. Conditional on choosing one parent at every reticulation, the network becomes a displayed tree. For $g = (g_i)_{i\in\mathcal{X}} \in \mathcal{G}^{\mathcal{X}}$,

$$\widehat{p}_N(g) = \begin{cases} \displaystyle\sum_{\sigma} w_\sigma \prod_{e\in T_\sigma} x_e^{\mathbf{1}[\oplus_{i\in A_e} g_i \neq 0]}, & \oplus_{i\in\mathcal{X}} g_i = 0, \\ 0, & \text{otherwise}, \end{cases} \tag{1}$$

where $A_e \mid A_e^c$ is the displayed-tree split. This is the usual group-based Fourier formula [7, 14].

The JC model depends only on the fixed mixed graph, not on its admissible rooted presentation. Indeed, rerooting preserves every displayed unrooted tree and its split multiset. Suppressing the temporary degree-two root artifact multiplies the adjacent JC edge multipliers; the effective product remains in $(0,1)$. Conversely, any effective multiplier $x \in (0,1)$ can be split inside the open domain, for example as $x = \sqrt{x}\sqrt{x}$, including when the root is placed adjacent to a retained reticulation edge. Reversing which incoming reticulation arc is called first only replaces λ by $1-\lambda$. These operations give open-domain parameterizations of the same Fourier formula (1). Write $\mathcal{M}_N^{\mathrm{JC}}$ for this open stochastic image and $\mathcal{V}_N^{\mathrm{JC}}$ for the complex Zariski closure.

Definition 2.4 (Observational relations)**.** *Let $d_N = \dim \mathcal{M}_N^{\mathrm{JC}}$, and let $(\mathcal{M}_N^{\mathrm{JC}})_{\mathrm{reg}}$ be the distributions having a parameter preimage at which the parameterization has Jacobian rank d_N. We write $N \bowtie_{\mathrm{JC}} N'$ if there is a point $p \in (\mathcal{M}_N^{\mathrm{JC}})_{\mathrm{reg}} \cap (\mathcal{M}_{N'}^{\mathrm{JC}})_{\mathrm{reg}}$ and a neighborhood of p in each regular image whose intersection has the full local dimension of both images. We write $N \preceq_{\mathrm{JC}} N'$ if there is a source-regular point $p \in (\mathcal{M}_N^{\mathrm{JC}})_{\mathrm{reg}} \cap \mathcal{M}_{N'}^{\mathrm{JC}}$ and a relatively open neighborhood U of p in $(\mathcal{M}_N^{\mathrm{JC}})_{\mathrm{reg}}$ such that $U \subseteq \mathcal{M}_{N'}^{\mathrm{JC}}$. The target dimension is allowed to be larger.*

These relations are distinct from equality of Zariski closures, equality of complete stochastic images, lower-dimensional intersection, and boundary-only intersection. We do not assume transitivity of the directed relation $\preceq_{\mathrm{JC}}$ a priori. Theorem 1.1 proves that, inside the stated class, it is symmetric and coincides with labelled isomorphism modulo T.

3 Primitive level-2 factors

Cut away the components beyond the boundary edges of one blob and retain their attachment sites as labelled ports. To form its homeomorphism core, temporarily remove the port arms, retain the local source and reticulation-sink events, and suppress every other ordinary bivalent side vertex, recording each removed labelled port in its directed order along the resulting core segment.

Lemma 3.1 (Root reduction)**.** *Every root-containing S_{TC} factor has the same open projective JC tensor germ as an incoming-port presentation at some real leaf-bearing boundary. For a comparison, the source and target incoming boundaries may be chosen independently.*

Proof. Starting at an admissible root, repeatedly choose a tree or leaf child. This reaches a labelled leaf without crossing a reticulation. Move the root down that ordinary path to the associated real boundary arm. Only ordinary arcs on the path reverse. At each path vertex the former tree parent becomes a tree child, while every off-path reticulation edge retains its arrowhead. If the reversal created a directed cycle, its first re-entry to the reversed path would give an internal path tree vertex a second old parent, contrary to binary indegree one; hence the rerooted graph is acyclic. Suppression of the old root joins its two old child arcs exactly as in the original one-step semi-directed reduction. If the off-path child is reticulate, its arrowhead is unchanged. Suppressing the newly inserted root therefore returns the same fixed mixed graph literally.

The chosen boundary side contains its labelled leaf. The complementary root side also contains a leaf: the other child of the old root cannot enter the chosen all-tree path, whose internal vertices have indegree one, and so must terminate at a leaf off that path. The path to the chosen leaf and one such off-path route are internally vertex-disjoint after the new root, again because no vertex of the all-tree path has a second entrance. Hence no proper descendant lies on every root-to-leaf path, and the new root is the LSA. The new site is admissible and is tree-child because membership in S_{TC} quantifies over every admissible rooting of this fixed mixed graph.

Reticulation parent choices are unchanged, up to replacing a bookkeeping probability λ by $1-\lambda$. Every switching therefore gives the same displayed unrooted tree, and JC reversibility preserves its split monomial. Splitting an arm multiplier x into $x_1x_2 = x$ or multiplying it after suppression stays inside $(0, 1)$. The only tensor change from moving within the arm is a positive incidence factor. The construction is existential for each factor and supplies no reason that two factors share one incoming boundary. □

Lemma 3.2 (Directed theta event placements)**.** *Let a nonroot binary level-2 factor have a theta underlying blob, one real incoming boundary, and two reticulations. After suppressing ordinary non-event vertices, its directed core is exactly one of $\theta_0, \theta_1, \theta_2, \theta_3$ in figure 2. No source or reticulation-sink placement is omitted.*

Proof. The unique incoming cut edge is ordinary: an edge entering a reticulation is not a bridge, because after deleting it its tail remains connected to the reticulation through the root, the reticulation's other parent, and the underlying graph of the rooted network. Its endpoint in the factor is consequently the unique local tree source S: the rooted bridge tree gives one parent bridge for the nonroot blob, and every other boundary is directed outward. A theta pole already has three incidences in the blob, so binary degree forbids the extra incoming boundary there; S is an internal point of one theta path.

At most one pole is reticulate. If both were, the path containing S points toward both poles and cannot contain either pole's unique outgoing side. The two outgoing sides must therefore lie on the other two event-free paths, forcing one complete path from the first pole to the second and another back from the second to the first. That is a directed cycle. Suppose first that one pole is reticulate. The other reticulation is an internal sink X, and the unique internal source S lies either on the same pole-to-pole path as X or on a different path. These two possibilities have only one acyclic order each. If the events lie on different paths, the S-path points away from S, the X-path points toward X, and the third path is forced from the tree pole to the reticulate pole; this is θ_0. If they share a path, the order from the tree pole to the reticulate pole is tree pole–S–X–reticulate pole. The reverse order would force the two event-free paths in opposite directions and create a directed cycle. Both event-free paths are therefore directed from the tree pole to the reticulate pole, giving θ_1.

Suppose instead that both poles are tree vertices. The two reticulations are internal sinks. They cannot lie on the same pole-to-pole path: along that path the two arrows entering each sink force a source between the sinks, while the two remaining event-free pole-to-pole paths would then each have to be directed from one tree pole to the other. They cannot have the same direction because the receiving pole would have two parents, and opposite directions form a directed cycle with the source-to-sink subpaths. Thus the sinks lie on distinct paths. The unique source lies either on the third path or shares a path with one sink. In the latter case acyclicity again fixes the source before the sink when read from the pole whose remaining two sides point outward. These are θ_2 and θ_3, respectively. Path reversal and interchange of undistinguished poles or sinks produce only the displayed symmetries, not additional event types. □

Here and below a *complete factor* is a ported blob factor with every physical cut-edge attachment retained as a labelled boundary port, including the child port below each path-sink reticulation.

Proposition 3.3 (Primitive-core theorem)**.** *Every nontrivial blob factor induced by a standard semi-directed network in S_{TC} has a cycle core or one of four directed theta cores. Every such complete factor is recovered uniquely by finite ordered words of port-bearing subdivisions on the directed core segments and by one real child port below every path-sink reticulation.*

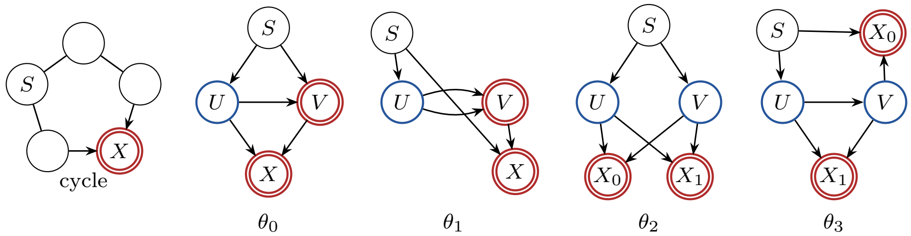


Figure 2: The primitive directed skeletons used by the local theorem: one cycle and four theta orientations. Parallel skeleton sides, visible in θ_1, must be subdivided by an occupied port before the fixed mixed graph is admitted as simple. Minimum strong repairs and arbitrary port subdivisions are added to these skeletons by the structural theorem.

Proof. For a root-containing factor first apply lemma 3.1; it therefore suffices to classify a nonroot factor with one real incoming boundary. Let t be the number of local tree vertices (including its endpoint source), r the number of local reticulations, $v = t + r$ the number of internal blob vertices, and e the number of blob edges after deleting the incoming and all outgoing port arms. Summing local indegrees, with the incoming boundary contributing the one omitted parent incidence, gives

$$e = t + 2r - 1 = v + r - 1.$$

Its cyclomatic number is therefore r. In the underlying blob every vertex has blob degree two or three: a port-bearing ordinary subdivision has blob degree two, even though it has a third external incidence before the port arm is removed. Since the blob is biconnected, every blob degree is at least two, and

$$\sum_{w \in V(B)} (\deg_B(w) - 2) = 2e - 2v = 2(r - 1). \tag{2}$$

For $r = 1$, every vertex has degree two, so B is a cycle. For $r = 2$, (2) gives exactly two degree-three branch vertices and all other vertices have degree two. Removing the two branch vertices leaves paths. Biconnectedness excludes a path returning to only one branch vertex, so the three incidences at either branch pair with the three incidences at the other. Thus the blob is the union of three internally vertex-disjoint paths between the two branches, namely a theta. Suppressing their ordinary degree-two non-event vertices gives the directed homeomorphism core, while the source and reticulation-sink events are retained and the remaining port-bearing subdivisions become ordered words on its segments.

The directed event placements are exactly the four cases of lemma 3.2. Reading port vertices in directed order on each segment recovers the words, and contracting those words recovers the unique core. □

The two underlying simple strict level-2 semi-directed generator shapes are also the two shapes in Huber et al. [12, Lemma 4.2 and Figure 8]: one has a reticulate pole and the other has two internal reticulation sinks. Our θ_0, θ_1 refine the first shape, and θ_2, θ_3 refine the second, according to the incoming source placement. The present ported theorem additionally records incoming roles, minimum strong repairs, labelled attachments, and their ordered subdivision words.

For reference, the segment templates and minimum no-omnian repairs are:

core	directed segments	minimum repairs
cycle	$S \to X, S \to X$	$\{0\}, \{1\}$
θ_0	$S \to U, S \to V, U \to X, V \to X, U \to V$	$\{2,3\}, \{3,4\}$
θ_1	$S \to U, S \to X, V \to X, U \to V, U \to V$	$\{2,3\}, \{2,4\}$
θ_2	$S \to U, S \to V, U \to X_0, V \to X_0, U \to X_1, V \to X_1$	$\{2,3\}, \{2,5\}, \{3,4\}, \{4,5\}$
θ_3	$S \to U, S \to X_0, V \to X_0, U \to X_1, V \to X_1, U \to V$	$\{2\}, \{4\}$

Within each row, segment indices start at 0 and follow the left-to-right order in the directed-segment column; the repair sets refer to those indices. Occupying one listed repair is necessary and sufficient because occupancy is monotone and the repairs respectively give a nonreticulate child to each offending tail, separate a reticulation child, or split a parallel core pair. This is the local form of the no-omnian criterion lemma 2.2; every path-sink child port is retained separately.

Proposition 3.4 (Triangles are automatically unique). *Every simple standard semi-directed network in S_{TC} has at most one triangle in each level-2 blob.*

Proof. A cycle has one simple cycle. A theta with path lengths (ℓ_1, ℓ_2, ℓ_3) has cycle lengths $\ell_i + \ell_j$. Two triangles force $(1,1,2)$ or $(1,2,2)$. The first has parallel pole edges. The second is $K_4 - e$. Two reticulations require four incoming arrowheads. A reticulation cannot tail one of these arrowheads, and lemma 2.2 allows each ordinary vertex to tail at most one. The mixed core has only two ordinary vertices, so it can supply at most two of the four required tails. Hence the simple mixed graph cannot belong to S_{TC}. □

Thus theorem 1.1 applies to all of S_{TC} at level two; no separate one-triangle-per-blob assumption is needed.

4 Pointwise recovery of cuts

For a split $A \mid A^c$, arrange the Fourier flattening by the total character $h = \oplus_{i \in A} g_i$. It has four character blocks.

Lemma 4.1 (Crossing-quartet reduction). *Let $A \mid A^c$ be a noncut split. A four-leaf restriction witnessing that failure can be chosen so that, after every selected-leaf-free bridge path is compressed, its central configuration is either one active component or one effective bridge joining two active three-boundary endpoint tensors.*

Proof. Color each labelled leaf by its side of the proposed split in the reduced bridge tree T. A singleton side is the pendant cut at that leaf, so a noncut split has at least two leaves of each color. Suppose first that some bridge of T has both colors on each side. Choose one leaf of each color on each side. In the minimal subtree spanning those four leaves, each side has a unique median at which its two colored branches separate from the central path. Every component strictly between the medians has only two selected boundaries and therefore contracts, in the selected Fourier tensor, to a positive arm or bridge multiplier. The result is one effective bridge joining the two median three-boundary tensors.

Otherwise no bridge has both colors on both sides. The convex hulls in T of the two color classes intersect—because a disjoint pair would be separated by a cut edge—but their intersection contains no edge. It is therefore one component v. Every incident branch of $T - v$ is monochromatic, and at least two branches have each color; with only one, its incident edge would realize the proposed split. Select one leaf from two branches of each color. The selected central configuration is the single active component v, while all selected-leaf-free paths again compress to positive arms. These two alternatives exhaust the reduced bridge tree. $\square$

Lemma 4.2 (Noncut-preserving word compression)**.** *Let H be a complete level-2 factor induced by a standard semi-directed network in S_{TC}. Let a two-color split of its labelled ports be noncut, with at least two labels of each color. There is a restriction to at most eight ports that retains the primitive core, one complete minimum strong repair, every path-sink child, at least two actual labels of each color, and a blob crossing the proposed split. In particular, the restricted split remains noncut.*

Proof. First retain a rigid support Q: every path-sink child and one occupied port on each segment of one minimum repair. By the table in section 3, $|Q| \leq 4$, and suppressing the unselected ordinary subdivisions leaves the entire primitive cycle or theta core. On each directed segment, read the two colors as a word and regard maximal constant subwords as monochromatic runs. For each color choose actual labels from at most two runs, prioritizing labels already in Q, until two labels of that color have been selected. If only one run is needed and its first chosen label is the sole selected label of that color, choose an actual adjacent label in the same run when one exists; otherwise choose an actual label from another run or segment. Such a label exists because each color occurs at least twice. No taxon is duplicated. This adds at most four labels to Q.

Deleting the other port arms and suppressing their now-bivalent attachment vertices does not delete a core edge, a repair segment, or a path-sink child. Hence the selected core is still one biconnected blob, and every selected label attaches to it by its own pendant arm. Any edge separating the two selected color classes would therefore have to lie inside that blob, but no blob edge is a bridge; a pendant arm separates only one selected label, whereas each color has at least two. Thus no bridge realizes the proposed split. The selected restriction is noncut and uses at most $4 + 4 = 8$ ports, below the ten-port bound used by the local certificates. $\square$

Theorem 4.3 (Pointwise cut theorem)**.** *At every point of the open JC model of a network in theorem 1.1,*

$$A \mid A^c \text{ is a cut split} \quad \Longleftrightarrow \quad \operatorname{rank} \operatorname{Flat}_{A|A^c}(p) \leq 4.$$

Proof. At a cut, each character block is a positive rank-one outer product of the two side tensors, with the bridge multiplier appearing in nonzero sectors. Hence the rank is at most four.

For the converse, a noncut has either one active central factor or a central bridge joining two active three-port endpoint tensors by lemma 4.1. In a one-active factor apply lemma 4.2. The four-active handoff in lemma 4.4 selects an actual quartet that remains wrong in one displayed switching, retains every other repair or sink role as a zero-character completion port, and supplies a strict block minor. Thus its Fourier flattening has rank at least five.

For the two-active case, number the two outer boundaries of the first endpoint by $1, 2$ and its central boundary by 3. After all selected-leaf- free serial paths have been contracted, let $u \in (0, 1]$ be the product of the complete effective edge class whose zero-sum signature is $\mathbf{1}[h_3 \neq 0]$; an empty serial class has $u = 1$, so $u \in (0, 1]$. If $\widetilde{P}$ is the physical endpoint tensor, define its central-normalized representative sectorwise by

$$P(g_1, g_2, h_3) = u^{-\mathbf{1}[h_3 \neq 0]} \widetilde{P}(g_1, g_2, h_3).$$

Define Q and $v \in (0, 1]$ analogously at the other endpoint. Term by term,

$$\widetilde{P}(\cdot, h)\, x^{\mathbf{1}[h \neq 0]} \widetilde{Q}(\cdot, h) = P(\cdot, h)\, (xuv)^{\mathbf{1}[h \neq 0]} Q(\cdot, h).$$

Thus the normalized effective central scale is $z = xuv \in (0, 1)$. In this chart write

$$a = P(1,1,0), \quad b = P(1,0,1), \quad c = P(0,1,1), \quad t = P(1,2,3)$$

and use capitals for the second endpoint. Set

$$\Delta = abc - t^2, \qquad \Gamma = a - bc.$$

Residual outer-arm scaling by (α, β) sends (a, b, c, t) to $(\alpha\beta a, \alpha b, \beta c, \alpha\beta t)$, so Δ scales by $(\alpha\beta)^2$ and Γ by $\alpha\beta$. An ordinary trivalent endpoint has $(a, b, c, t) = (\alpha\beta, \alpha, \beta, \alpha\beta)$ and hence $\Delta = \Gamma = 0$. Every allowed endpoint satisfies the invariant dichotomy

$$\Delta > 0 \quad \text{or} \quad \Delta = 0 \text{ and } \Gamma \geq 0. \tag{3}$$

Index the proposed-split flattening by rows (g_1, g_3) and columns (g_2, g_4); a block consists of pairs with $g_1 \oplus g_3 = g_2 \oplus g_4$. After central normalization and division by a common positive outer row/column monomial, the following ordered minors are exactly the displayed polynomials:

minor	block total	ordered rows	ordered columns	normalized determinant
f_1	0	$(0,0),(1,1)$	$(0,0),(1,1)$	$aA - z^2bcBC$
f_2	0	$(0,0),(1,1)$	$(0,0),(2,2)$	$zTt - z^2bcBC$
f_3	1	$(0,1),(1,0)$	$(0,1),(2,3)$	$zC(At - zTbc)$
f_4	1	$(0,1),(1,0)$	$(1,0),(2,3)$	$zc(zBCt - Ta)$

All four character blocks contain positive entries, so each has rank at least one. If the full flattening had rank at most four, the four block ranks would each equal one; in particular $f_1 = f_2 = f_3 = f_4 = 0$. The exact identities

$$\begin{aligned} aA - zTt &= f_1 - f_2, \\ z^2CT\Delta &= zCt(f_1 - f_2) - af_3, \\ z^2ct\Delta' &= Af_4 + zcT(f_1 - f_2), \qquad \Delta' = ABC - T^2, \end{aligned}$$

then force $\Delta = \Delta' = 0$. By (3), $aA \geq bcBC > z^2bcBC$, contradicting $f_1 = 0$.

Marginalizing to the selected quartet sets every omitted leaf character to zero, so its Fourier flattening is a submatrix of the full flattening. In either central case at least one quartet character block has rank at least two. All four blocks have rank at least one because every open JC Fourier entry is positive. Hence the quartet, and therefore the full Fourier flattening, has rank at least five. Fourier transformation acts by invertible row and column changes of basis, so the ordinary pattern flattening has the same rank.

□

Lemma 4.4 (Exact cut certificate)**.** *The endpoint dichotomy* (3) *and every one-active case used in theorem 4.3 have graph-derived exact certificates. The separately implemented replay contains* 67 *endpoint types with* $\Delta > 0$*,* 2 *with* $\Delta = 0$ *and* $\Gamma > 0$*,* 7 *nonordinary endpoint types with* $\Delta = \Gamma = 0$*, the ordinary endpoint with* $\Delta = \Gamma = 0$*, and* 204 *strict one-active minors. All inequalities hold throughout the complete open parameter cube.*

Proof. For every primitive completion the verifier enumerates the displayed switchings, computes descendant masks, forms the exact Fourier tensor, and pulls back Δ, Γ or the selected flattening minor. For an endpoint, the complete central singleton-signature edge class is first identified and its effective product is set to one; this is exactly the sectorwise normalization in the proof of theorem 4.3. Each nonzero pullback is factored or converted to the Bernstein basis on the unit cube; all coefficients required for strict sign have one sign. A second implementation reconstructs all graph-to-polynomial assignments without importing the producer. The full normalized records and all 204 minors agree.

Here is the exact handoff from the at-most-eight-port compression to those four-active-port records. On each directed segment, if the colour word has three monochromatic runs c, d, c, then in every switching the path between the two outer c attachments contains the middle d attachment. No tree edge can separate the complete colour classes: a segment edge separates the outer c ports or keeps the middle d with them, a pendant edge isolates one port, and a second d port exists elsewhere. Such a word is therefore a direct all-switching obstruction.

Otherwise every segment has at most two runs. Restrict to one actual port per run, preserving every occupied segment and fixed extra role. Each word is then one of

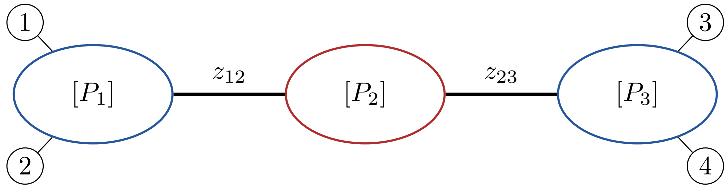
$$(), \quad (0), \quad (1), \quad (0,1), \quad (1,0).$$

If one colour now has only one representative, all its original occurrences lay in that run; retain a second adjacent actual port there. This is the singleton-doubled palette case. Segment occupancy, the minimum repair, and standard-strong validity are unchanged. Since every displayed split remains displayed after leaf restriction, an all-switching survivor before reduction would survive in this finite palette.

A standalone combinatorial implementation exhausts all 808,642 balanced four-through-eight-port binary word distributions and verifies this partition. A separately written graph/switching implementation and the producer independently check the complete direct and singleton-doubled palette; both have zero survivors. Thus no balanced noncut colouring is displayed by every switching. Choose a switching that does not display the selected split. The ordinary tree quartet criterion then gives two actual labels of each colour whose induced quartet still fails to display that split. Every other selected repair or path-sink role is retained with character zero, so it remains a physical completion role rather than being deleted. The four-port compiler enumerates exactly these 72 active-labelled tensors. It skips the 12 directions displayed by every switching and gives one strict minor for each of the other 204 directions. That quartet minor is a submatrix of the compressed and full flattenings. This is a finite exact proof because proposition 3.3 and lemma 4.2 exhaust the primitive completions and compressed colourings; the counts are checksums, not the premise. ☐

Corollary 4.5 (Bridge-tree equality under one-sided containment)**.** *If $N \preceq_{\mathrm{JC}} N'$, then N and N' have the same labelled cut splits and the same reduced bridge tree, including ordinary/nontrivial component decoration.*

Proof. On the common source-open set, a source cut cannot be a target noncut by theorem 4.3. Conversely, the rank equations for a target cut hold on the same set and cannot hold at any source noncut point. Hence both cut sets are equal. Their compatible split system reconstructs the reduced labelled tree. For a degree-three component, $\Delta = abc - t^2$ is zero on an ordinary median and strictly positive on an open three-sunlet. A strong theta has at least four boundaries. Thus component decorations agree as well. ☐



$P_v \mapsto P_v \prod_{e \ni v} a_{v,e}^{\mathbf{1}[h_e \neq 0]}, \quad x_e \mapsto x_e/(a_{u,e}a_{v,e})$

Figure 3: Bridge-tree peeling recovers a projective local tensor orbit $[P_v]$ at each retained component and normalized effective bridge scales z_e. The original physical bridge multiplier is not an intrinsic coordinate; the exact fiber is the full incidence-scaling orbit intersected with the positive JC domain.

5 Projective bridge peeling

Let T be the recovered bridge tree. A component tensor P_v is supported on assignments satisfying local character conservation and is normalized by

$$P_v(0, \dots, 0) = 1.$$

For a bridge $e = uv$ with multiplier x_e, the global Fourier tensor is

$$\Gamma(P, x)(g) = \prod_v P_v(g_v, h_v) \prod_{e \in E(T)} x_e^{\mathbf{1}[h_e \neq 0]}. \tag{4}$$

The *positive incidence-saturated tensor locus* is the saturation under (5) of the positive, conservation-supported, JC-symmetric, componentwise normalized tensor locus together with positive bridge scales. A physical network model is the semialgebraic subset cut out by its local realizability and open parameter inequalities.

Theorem 5.1 (Exact positive bridge fiber)**.** *On the positive incidence-saturated JC-symmetric tensor locus, the complete fiber of* (4) *is the incidence action*

$$P_v \longmapsto P_v \prod_{e \ni v} a_{v,e}^{\mathbf{1}[h_e \neq 0]}, \qquad x_e \longmapsto \frac{x_e}{a_{u,e}a_{v,e}}, \tag{5}$$

with all $a_{v,e} > 0$. Restricted to physical network realizations, the fiber is the intersection of this ambient incidence orbit with the complete *physical realization locus: every local arm multiplier, every bridge multiplier, and every inheritance probability must remain in its open JC domain. Every retained component orbit admits a positive real-analytic local slice in the incidence-saturated tensor space. The sliced component tensors and normalized effective bridge scales are intrinsic functions of the distribution; the slice tensors need not themselves be physical local network tensors.*

Proof. Cut T at one bridge and fix a character sector h. The corresponding flattening block is a positive rank-one matrix. Uniqueness of positive rank-one factors gives a sector scale $c_e(h)$ on one side and its reciprocal on the other. Since both normalized subtree contractions have all-zero coordinate one, $c_e(0) = 1$. The six automorphisms of $\mathcal{G}$ act transitively on the three nonzero characters, and both factorizations are JC-symmetric, so $c_e(h)$ has one common value for $h \neq 0$.

Root the bridge tree temporarily and peel its leaf components. Expanding a peeled subtree contraction toward the root gives, at an unpeeled component v,

$$\frac{Q_v}{P_v} = c_e(h_e) \prod_{f = vw} \left(c_f(h_f)^{-1} (x_f/y_f)^{\mathbf{1}[h_f \neq 0]} \right),$$

where e is the parent incidence and the product is over child incidences. Assigning the parent and child factors separately defines one positive number $a_{v,f}$ at every incidence and yields exactly (5). Induction removes every component, so there is no additional vertex-coupled freedom. Direct cancellation proves the converse and hence the complete positive fiber.

For a component carrying a physical leaf block, one-character anchor ratios normalize the incidences independently; their log-exponent matrix is the identity. An unmarked retained component has degree $d \geq 3$. Its pair anchors

$$(1,2),(1,3),(2,3),(1,4),\ldots,(1,d)$$

have a log-exponent matrix whose first three rows have determinant -2 and whose later rows each introduce one new coordinate. It has rank d, and positive square-root formulas give a unique real-analytic normalizer. Thus the incidence action is free on every retained component. The excluded unmarked degree-two stabilizer would require the two-port $K_4 - e$ theta, which violates the strong criterion; ordinary unmarked components and cycle components have degree at least three.

These normalizers define analytic slices in the incidence-saturated positive tensor space. Restricting an orbit to physical realizations means intersecting it with all open inequalities defining realizable component tensors and bridge multipliers; the displayed inequality $0 < x_e/(a_{u,e}a_{v,e}) < 1$ is necessary but not sufficient by itself. Arbitrary positive scales are not asserted to remain physical. Finally, T is a tree, so peeling reaches every incidence and no gauge can circulate as holonomy around a cycle. □

Remark 5.2. The old reciprocal-only transformation is false. For example, $(y,z,x) = (1/2,1/2,1/2)$ and $(3/5,3/5,25/72)$ have the same observable product $xyz = 1/8$ but are not related by reciprocal endpoint scaling. The theorem recovers normalized effective scales, not physical bridge multipliers.

Lemma 5.3 (Local product chart)**.** *On a positive regular physical locus, after choosing the incidence normalizers of theorem 5.1, contraction along the recovered bridge tree is a real-analytic local diffeomorphism*

$$\prod_v (\textit{projective local image germ at } v) \times \prod_{e\in E(T)} I_e \longrightarrow \textit{the global model germ},$$

where every I_e is an interval of normalized effective bridge scales. This statement concerns image germs on the physical realization locus, not the original physical edge parameters.

Proof. Positive rank-one cut factorization extracts both side factors analytically, up to one positive scalar in each nonzero character sector. JC symmetry identifies the three nonzero sector scalars, and the full-rank anchor systems in theorem 5.1 select one analytic incidence representative of each projective component orbit. The residual scalar in each cut is the normalized effective bridge scale. Peeling the bridge tree recursively therefore defines an analytic extraction map. Its composition with (4) is the identity on the product coordinates, while the exact fiber theorem shows that contraction after extraction returns the original tensor. Thus contraction and extraction are local inverses. The bridge graph is a tree, so the recursive normalizations cannot acquire holonomy. Shrinking the chart keeps it inside the open physical realization locus on which it was constructed. More explicitly, at each regular physical parameter point the constant-rank theorem gives a local analytic section onto the chosen smooth local image branch associated with that preimage. Composing this section with the incidence normalizer makes the normalized projective tensor coordinates analytic on a physical local image germ. The ambient normalized tensor locus, the physical realization locus, its projective image, and the chosen incidence slice are therefore distinct objects; in particular, a slice representative need not itself be physical. □

Lemma 5.4 (Semialgebraic interior and finite covers)**.** *Let M be a smooth d-dimensional semialgebraic manifold and $S \subseteq M$ semialgebraic. Then $\dim S = d$ if and only if S has nonempty relative interior in M. Consequently, if a nonempty relatively open subset $U \subseteq M$ is covered by finitely many semialgebraic subsets $A_1, \dots, A_m \subseteq M$, then, for some i, the intersection $A_i \cap U$ contains a nonempty relatively open subset of U.*

Proof. Choose a finite semialgebraic atlas of M. Semialgebraic bijections preserve dimension by Bochnak–Coste–Roy, Theorem 2.8.8. If S has nonempty relative interior, one chart image of S contains a nonempty open semialgebraic subset of $\mathbb{R}^d$, which has dimension d by Proposition 2.8.4.

Conversely, suppose that S has empty relative interior. In any chart with open image $O \subseteq \mathbb{R}^d$, let S' be the chart image of the corresponding part of S and put $C = O \setminus S'$. Then C is dense in O. Since $O \subseteq \overline{C}$, Propositions 2.8.2 and 2.8.4 give $\dim C = d$. Proposition 2.8.13 gives

$$\dim(\overline{C} \setminus C) < d,$$

and $S' \subseteq \overline{C} \setminus C$. Hence every chart part of S has dimension less than d. Proposition 2.8.5(i), applied to the finite atlas, gives $\dim S < d$.

For the finite-cover conclusion, choose a nonempty relatively open semialgebraic neighborhood $V \subseteq U$. Then

$$V = \bigcup_{i=1}^{m} (V \cap A_i)$$

for the finitely many covering sets $A_1, \dots, A_m$, and $\dim V = d$. Proposition 2.8.5(i) makes some $V \cap A_i$ d-dimensional. By the first part, it has nonempty relative interior in M; because it lies in U, it contains a nonempty relatively open subset of U. See [2, Theorem 2.8.8 and Propositions 2.8.2, 2.8.4, 2.8.5(i), and 2.8.13]. □

Proposition 5.5 (Localization and no compensation)**.** *If $N \preceq_{\mathrm{JC}} N'$, then for every pair of corresponding nontrivial factors H, H' there is a source-relative full-dimensional regular containment of the projective local JC germ of H in one projective incoming/completion germ of H'. Parameters in other blobs cannot compensate for a local separator.*

Proof. Let p and U witness $N \preceq_{\mathrm{JC}} N'$, and choose a source parameter preimage of p having rank d_N. A nonzero maximal minor remains nonzero near that preimage, so the constant-rank theorem supplies its smooth d_N-dimensional local source-image branch. The smooth locus of the irreducible source model closure is dense. After moving the base point inside the same source-open containment and shrinking, we may therefore take U to be a nonempty relatively open subgerm of one smooth source branch, still contained in the target model. No target parameter section is chosen.

By lemma 5.3, in source slice coordinates this germ contains a smaller product box of local tensor germs and effective bridge intervals. Vary one focal factor and fix the other coordinates. Every target realization of the same distribution has the same intrinsically extracted focal projective orbit by theorem 5.1. The finitely many target incoming and completion types cover the focal box. Lemma 5.4 therefore places a source-relative full-dimensional open *subgerm* into one type. No continuous choice of target parameters is made. Since extraction is an intrinsic function of the distribution, a distant factor cannot change the focal orbit or cancel a projective invariant. □

6 The complete local theorem

For a port tensor $P(g_1, \ldots, g_k)$, quotient by the positive action

$$P(g) \longmapsto P(g) \prod_i a_i^{\mathbf{1}[g_i \neq 0]}.$$

Let $\mathbb{P}\mathcal{M}_H$ denote the resulting open projective JC model of a complete ported factor H. Put $d_H = \dim \mathbb{P}\mathcal{M}_H$, and let $(\mathbb{P}\mathcal{M}_H)_{\mathrm{reg}}$ be the set of projective tensors having a physical parameter preimage at which the projective parameterization has rank d_H. For complete ported factors H, H' on the same labelled boundary set, write

$$\mathbb{P}\mathcal{M}_H \preceq_{\mathrm{proj}} \mathbb{P}\mathcal{M}_{H'}$$

if there are a point $P \in (\mathbb{P}\mathcal{M}_H)_{\mathrm{reg}} \cap \mathbb{P}\mathcal{M}_{H'}$ and a relatively open neighborhood U of P in $(\mathbb{P}\mathcal{M}_H)_{\mathrm{reg}}$ such that $U \subseteq \mathbb{P}\mathcal{M}_{H'}$. The target dimension is allowed to be larger.

Theorem 6.1 (Local blob-containment theorem)**.** *Let H, H' be complete nontrivial factors on the same labelled physical boundary set, induced by standard semi-directed S_{TC} level-2 networks. Choose admissible incoming presentations independently. Then*

$$\mathbb{P}\mathcal{M}_H \preceq_{\mathrm{proj}} \mathbb{P}\mathcal{M}_{H'} \quad \iff \quad H \text{ and } H' \text{ are labelled-isomorphic or } \mathsf{T}\text{-related.}$$

There is no proper one-sided projective local containment.

The necessity proof is finite after a structural support reduction. We give its mathematical completeness argument before stating the exact computer-assisted lemma.

Lemma 6.2 (Marginal open image)**.** *Let R be any rigid-support, restoration-prefix, support-plus-one, or support-plus-two restriction used below. On a dense regular locus of the source model, R has the generic rank of the selected source model. Consequently the image under R of every nonempty source-relative open containment contains a nonempty relatively open subset of that selected model.*

Proof. Work on a smooth semialgebraic source-image stratum. For each switching, the restriction sets omitted leaf characters to zero. Group physical edges whose complete switching-signature rows have the same zero-sum JC indicator signature—that is, they induce the same JC exponent on every zero-sum assignment of the retained ports. This normalization identifies, in particular, a selected split mask with its complement; unnormalized rooted descendant-mask rows need not be equal. Discard the all-zero signature, whose physical parameters are tensor-invisible after restriction. Within every remaining equivalence class the selected parameter is

$$(x_1, \ldots, x_s) \longmapsto y = x_1 \cdots x_s, \qquad dy = \sum_i \left(\prod_{j \neq i} x_j \right) dx_i.$$

Every coefficient is positive on $(0,1)^s$, so this product map is a submersion. Distinct normalized indicator signatures give the distinct selected effective edge coordinates; inheritance coordinates are retained, with a possible parent flip $\lambda \mapsto 1 - \lambda$. The resulting descriptor map from full to selected physical parameters is therefore onto and submersive.

Choose a regular parameter point of the selected model and lift it through the descriptor, denoted δ_R. The selected-rank minor and a generic full-model rank minor are nonzero polynomials on the irreducible full parameter space: the first is nonzero because of that lift and the second by the definition of full model dimension. The preimage of the smooth locus of the irreducible full-model variety is also a nonempty Zariski-open set. These three open sets therefore meet. Choose θ in their intersection, leaving omitted parameters in their open intervals, and let M_{full} be the smooth source-image stratum through $\phi_{\text{full}}(\theta)$. The chain rule for $R \circ \phi_{\text{full}} = \phi_{\text{selected}} \circ \delta_R$ gives the required rank on parameter space. At this full-model regular point,

$$T_{\phi_{\text{full}}(\theta)} M_{\text{full}} = \operatorname{Im} D\phi_{\text{full}}(\theta).$$

Consequently the restriction of DR to this image tangent space has the same selected-model rank. The generic-rank theorem applied to the algebraic restriction of R to the smooth full-model locus now makes its rank-drop set a proper algebraic-minor locus there, hence a semialgebraic set with empty relative interior. Every nonempty source-relative open containment meets the dense maximal-rank locus. At such a point the constant-rank theorem makes R an open map onto a relative neighborhood in the selected smooth image. Semialgebraicity and the asserted density follow from Tarski–Seidenberg and the dimension results in Bochnak–Coste–Roy [2, Theorem 2.2.1, applied iteratively, and Propositions 2.8.2, 2.8.4, and 2.8.13]. □

6.1 Rigid supports and exact completions

Choose every reticulation-sink child and one labelled port on each segment of one minimum repair. The resulting support Q retains the primitive core and is pointwise rigid. Its outgoing sizes are

$$2 \quad (\text{cycle}), \qquad 3 \quad (\theta_0, \theta_1, \theta_3), \qquad 4 \quad (\theta_2).$$

The minimal two-outgoing cycle is handled directly by

$$\mathcal{I}_{\text{tri}} := q_{011} q_{101} q_{110} - q_{123}^2, \tag{6}$$

which vanishes on an ordinary median and is strictly positive on every open three-sunlet. This trinet polynomial and its tree–reticulation sign dichotomy are due to Englander et al. [6, Proposition 2.15]. In their collapsed JC notation, the orbit coordinate denoted here by q_{123} is denoted q_{111}. Our use here is its integration into the complete directed level-2 atlas and the restored-support argument. The three sunlet orientations are exactly T.

Marginalize a hypothetical full containment to Q. By lemma 6.2, this produces a source-relative open containment in the selected source model. On every target core, distribute the selected ordinary labels in order among its directed segments. Its structural incoming boundary may be selected or have character zero. Restore every omitted reticulation-sink child and every empty segment of a chosen minimum repair by one zero-character dummy. Any further omitted ordinary port lies in a serial class and contributes only through a positive product. Conversely, contracting unselected ordinary subdivisions in a full strong target produces exactly one such raw record. Hence the completion grammar is onto every selected target tensor.

Let $\{Y_\tau\}_{\tau \in \mathcal{T}}$ be the selected projective images of the finitely many target incoming/completion types produced by this grammar. The source-relative open selected germ U is contained in $\bigcup_{\tau \in \mathcal{T}} Y_\tau$, and every Y_τ is semialgebraic. By lemma 5.4, some Y_τ contains a nonempty relatively open full-dimensional subgerm $U_\tau \subseteq U$. Fix that decorated directed relation before applying the bounded theorem below. Thus no continuous choice of target parameters or completion type is assumed.

For each displayed switching and physical edge, record the descendant mask on the selected ports and normalize its complete row by its zero-sum JC indicator signature, including split complements.

Edges in one normalized class occur in every Fourier monomial with equal exponents and are replaced by their product; an all-zero class is omitted as tensor-invisible. The serial products and possible parent complements are exactly the descriptor maps in lemma 6.2. Thus the descriptor gives the exact positive selected tensor image, not a boundary specialization.

On every ordered four-port marginal the certificate evaluates the S_4-orbit of seven explicit integer JC invariant templates. The templates and all 84 distinct transported polynomials are in the machine-readable appendix. Every polynomial is multihomogeneous in the port arms, so its zero or strict-sign status descends to the projective quotient. If an invariant vanishes identically on the target but not on the source, source-full containment would put an open part of an irreducible source model in a proper algebraic subset. This gives the exact necessary filter

$$S(\text{source}) \subseteq S(\text{target}),$$

where S records nonidentically-zero pullbacks. A modular nonzero value is used only as an exact nonvanishing witness; every modular zero is expanded over $\mathbb{Z}$.

6.2 The bounded exact theorem

Proof boundary. The human proof supplies primitive-core exhaustiveness, the strong repairs, rigid supports, marginal open image, restoration, coherent probes, root reduction, projective localization, and physical gluing. The sole finite machine-certified dependency is the following bounded directed-relation statement; every polynomial used in it is regenerated from its two decorated graphs. The complete canonical relation universe, raw-to-canonical transports, per-relation exact certificates, restoration and probe records, and primary and separately implemented replay verifiers are archived in the curated proof-certificate bundle [13]. Its `ACTIVE_MANIFEST.json` authenticates every proof-payload file, `SHA256SUMS` is checked as its exact projection, and the external archive SHA-256 authenticates the complete compressed object. The file `atlas/ATLAS_EVIDENCE_BINDINGS.jsonl.gz` binds each decorated relation to its exact evidence and replay program. The command `bash verify.sh regenerate-all` reconstructs the complete frozen atlas and those row-level bindings from the primitive inputs.

Theorem 6.3 (Finite decorated-relation theorem)**.** *The certificate assigns every canonical decorated directed relation in the certified bounded universe to exactly one of the following categories, and the stated certificate property holds in that category:*

A. *a regenerated exact pullback or flattening minor strictly separates the relation on the open cube;*
B. *it belongs to a verified fixed-full restoration forest whose terminal paths are separated or labelled-isomorphic modulo* T*; or*
C. *it is directly labelled-isomorphic or* T*-related.*

No nonisomorphic, non-T *relation survives in either containment direction.*

Proof. The theorem object is the coloured disjoint union of the source and target mixed graphs together with side colours, independently distinguished incoming boundaries, all physical port labels and matches, omitted-role placeholders, and source-to-target direction. Canonicalization retains explicit vertex, edge, inheritance, label, and Fourier-coordinate transports. Thus two relations sharing an unlabelled target cannot be merged by a target hash.

For the three-outgoing supports, independent generation gives 10,466 canonical relations from 10,826 raw presentations. For the four-outgoing θ_2 gate, expanding every surviving provenance record gives 192 raw directed presentations. These numbers identify the two certified finite universes; the category, restoration, probe, and mutation checksum tables are in the supplement.

For every strict record the verifier reconstructs both rooted graphs, enumerates displayed switchings and descendant masks, forms the complete exact JC tensors, regenerates the chosen pullback, and proves its open-cube sign by exact factors or rational Bernstein coefficients. For every equality terminal, independent standard mixed-graph canonicalization accepts only labelled isomorphism or the quotient forgetting the arrowheads of one triangle. Complete normalized primary and clean-room records agree, including every raw-to-canonical transport and every restoration-parent edge.

The finite universe is exhaustive by proposition 3.3, the onto completion grammar, every relative port permutation, and both incoming modes. The integer counts in the supplement are reproducibility checksums. Mutation suites reject relation deletion or duplication, changed port correspondences, collapsed source embeddings, swapped separators, reversed direction, changed incoming roles, broken restoration parents, and valid polynomials attached to the wrong graph. Therefore the exact finite alternatives (a)–(c) hold. □

6.3 Restoration and arbitrary words

The hard cover never infers a larger marginal from containment of a smaller one. Fix the complete relation first. Let Q_s, Q_t be source and target rigid supports, and set $A = Q_s \cup Q_t$. Every target dummy role is filled by its actual physical label in the full comparison. For each prefix D_j of those labels,

$$\mathbb{P}\mathcal{M}_{H|_{Q_s \cup D_j}} \preceq_{\mathrm{proj}} \mathbb{P}\mathcal{M}_{H'|_{Q_s \cup D_j}}$$

is a direct marginal of the original full containment. By lemma 6.2, each prefix contains a source-relative open selected-model subgerm. Hence the actual restoration path cannot meet a separating certificate and must terminate in a common core-retaining anchor A modulo T.

The 62 direct residual anchors require a separate base case: they have four selected ports, while every restoration terminal has at least five. Each has one canonical transport (34 labelled isomorphisms and 28 ordinary T relations). Independently inserting one new labelled port on every ordered source–target pair of admissible blob arcs gives all 2,642 $A \cup \{p\}$ relations. Exactly 314 remain isomorphic or T-related. Inserting a second port above precisely those parents gives all 18,224 $A \cup \{p, q\}$ relations, of which 2,032 remain isomorphic or T-related. Every other relation has a regenerated exact JC separator, and every surviving child transport restricts its unique parent transport. Thus direct anchors and restoration terminals satisfy the same coherence hypothesis below.

Lemma 6.4 (Coherent-probe theorem)**.** *After fixing the one anchor transport on A, the restrictions $A \cup \{p\}$ locate every additional port, and $A \cup \{p, q\}$ determines the order of each pair on one segment. All probes assemble to one complete port-word isomorphism modulo one coherent ordinary triangle redirection. At most ten tensor ports are required.*

Proof. The anchor is pointwise rigid on both sides. Therefore every one-port probe restricts the same core map and fixes the segment and interval containing p. Every two-port probe restricts its exact one-port parent and determines the order of p, q when they lie in one interval. These comparisons are restrictions of actual finite total words and assemble uniquely. If a probe subdivides a triangle edge, its restriction has no triangle and forces literal orientation agreement; otherwise the same unique triangle persists, so all probes have one common T choice.

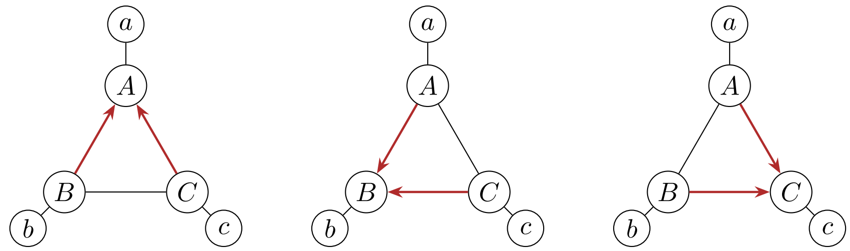


Figure 4: Ordinary triangle redirection T. The labelled underlying graph and all external attachments are fixed; only the reticulation vertex and the two incoming arrowheads inside the triangle change. The three local JC models share a regular full-dimensional germ, not necessarily their entire open stochastic images.

The source and target rigid supports together use at most eight tensor ports in the attained relations; adding two probes gives ten. Independent compact path-bound and direct-anchor replays cover every permitted probe relation; their exact checksum counts are in the supplement. Adversarial word tests include empty segments, repeated subdivisions, symmetric labels, reversed orders, and inconsistent T choices. Every mutation is rejected. □

The left-to-right implication of theorem 6.1 now follows from theorem 6.3 and lemma 6.4.

6.4 The ordinary triangle converse

Lemma 6.5 (Ordinary triangle germ)**.** *The three labelled orientations of a port-labelled three-sunlet share a four-dimensional regular open JC tensor germ in the strict parameter domain. The same is true after inserting the triangle in any unchanged corresponding local context.*

Proof. At an exact rational point, direct displayed-tree summation gives equality of all 64 Fourier entries for all three orientations. Their four nonconstant JC orbit coordinates are

$$(q_{12}, q_{13}, q_{23}, q_{123}) = \left(\delta^2, \delta^2, \delta^2, \frac{4}{5}\delta^3\right), \qquad \delta = 2^{-10}.$$

Each physical orientation has a nonzero rank-four Jacobian minor. Four is also the normalized three-port ambient dimension, so every orientation maps submersively onto a neighborhood of the common tensor. Positive port-arm scaling preserves a common projective germ.

We now separate the one projective tensor direction from the three incidence directions, and put $\epsilon(h) = \mathbf{1}[h \neq 0]$. Write $F_i : \Theta_i \to \mathcal{O}$ for the physical triangle parameterization into a common positive four-dimensional tensor neighborhood $\mathcal{O}$ of the certified tensor. Fix its three pair coordinates $q^*_{12}, q^*_{13}, q^*_{23}$. Every $Q \in \mathcal{O}$ has a unique analytic decomposition

$$Q(\mathbf{h}) = \overline{Q}_u(\mathbf{h}) \prod_{j=1}^{3} a_j(Q)^{\epsilon(h_j)}, \qquad q_{12}(\overline{Q}_u) = q^*_{12}, \quad q_{13}(\overline{Q}_u) = q^*_{13}, \quad q_{23}(\overline{Q}_u) = q^*_{23}.$$

Indeed, putting $b_{jk} = q_{jk}(Q)/q^*_{jk}$ gives

$$a_1 = \sqrt{b_{12}b_{13}/b_{23}}, \qquad a_2 = \sqrt{b_{12}b_{23}/b_{13}}, \qquad a_3 = \sqrt{b_{13}b_{23}/b_{12}}.$$

Thus the incidence normalizer is a local analytic diffeomorphism $\mathcal{O} \simeq \mathcal{Q} \times \mathcal{A}$, where $\mathcal{Q}$ is the one-dimensional representative slice through the common tensor and $\mathcal{A} \subset \mathbb{R}^3_{>0}$ records the incidence scales. Equivalently, $q_{123}^2/(q_{12}q_{13}q_{23})$ is a local coordinate on the quotient.

Let $f_i : \Theta_i \to \mathcal{Q}$ be F_i followed by projection of this normalizer to $\mathcal{Q}$. Since F_i has rank four and the normalizer is a local diffeomorphism, df_i has rank one. After shrinking about the certified point, the constant-rank theorem gives a common open interval $U \subset \mathcal{Q}$ and analytic sections $s_i : U \to \Theta_i$. They need not realize the slice representative itself: for positive analytic functions $a_{ij}(u)$,

$$F_i(s_i(u))(\mathbf{h}) = \overline{Q}_u(\mathbf{h}) \prod_{j=1}^{3} a_{ij}(u)^{\epsilon(h_j)}.$$

To make contextual substitution explicit, cut the three attachment edges between the triangle and its unchanged context. Shrink U so that the a_{ij} and their reciprocals are bounded. Choose common effective boundary scales $z_j > 0$ sufficiently small and set

$$r_{ij}(u, \mathbf{z}) = \frac{z_j}{a_{ij}(u)} \in (0, 1).$$

Then the physical boundary arms give the termwise identity

$$F_i(s_i(u))(\mathbf{h}) \prod_{j=1}^{3} r_{ij}(u, \mathbf{z})^{\epsilon(h_j)} = \overline{Q}_u(\mathbf{h}) \prod_{j=1}^{3} z_j^{\epsilon(h_j)}.$$

If $C(g; \mathbf{h}, c)$ is the arbitrary multi-boundary Fourier tensor of the unchanged context, all orientations therefore feed into the same multilinear contraction

$$\Phi(u, c, \mathbf{z})(g) = \sum_{\mathbf{h} \in \mathcal{G}^3} C(g; \mathbf{h}, c) \overline{Q}_u(\mathbf{h}) \prod_{j=1}^{3} z_j^{\epsilon(h_j)}.$$

The context tensor may reconnect two triangle terminals inside one theta factor; no tensor-product independence is assumed.

Choose a connected positive chart $\mathcal{C}$ for $(c, \mathbf{z})$ and let d be the generic rank of this common map on $U \times \mathcal{C}$. The one coordinate u and three effective boundary scales $\mathbf{z}$ retain all four local triangle-tensor directions. By definition, some d-minor of $D\Phi$ is not identically zero there. Its nonvanishing locus is dense and open, so choose $(u_0, c_0, \mathbf{z}_0) \in U \times \mathcal{C}$ where the minor is nonzero. The sections s_i together with the physical arms r_{ij} define analytic physical sections $\sigma_i : U \times \mathcal{C} \to \Theta_i^{\text{context}}$ satisfying $\phi_i \circ \sigma_i = \Phi$. Hence the physical contextual map ϕ_i has rank at least d there.

For the reverse inequality, take an arbitrary nearby physical parameter point of orientation i and apply the explicit three-leg normalizer above to its triangle tensor. If its incidence coordinates are a_j, and its three physical boundary multipliers are r_j, define $z_j = a_j r_j$. Together with the unchanged context coordinates this is a direct analytic map $\pi_i : \Theta_i^{\text{context}} \to U \times \mathcal{C}$ for which $\phi_i = \Phi \circ \pi_i$, as the displayed multilinear identity verifies termwise. No bridge-cut extraction, and in particular no application of lemma 5.3, is used here: two of the three terminals may reconnect inside one theta factor. Thus $\operatorname{rank} D\phi_i \leq d$. The two inequalities give equality. Shrinking once more, $\Phi(U \times \mathcal{C})$ is therefore a common d-dimensional regular germ equal to the full local model dimension on every orientation. This is a forward contraction and an incidence normalization, not an attempted inverse factorization of a theta into two hidden-pole pieces. □

Labelled isomorphism is immediate, and lemma 6.5 proves the converse direction of theorem 6.1. Notice that the necessity argument classified the complete theta tensor directly; it did not treat a triangle-bearing theta as a bridge gluing of a sunlet and its complementary path.

Lemma 6.6 (Simultaneous physical gluing)**.** *Let two networks have the same bridge tree and finitely many corresponding local factors, each pair labelled-isomorphic or* T*-related. Their common projective tensor germs admit physical analytic sections on both networks. After shrinking, common effective bridge scales can be chosen so that every physical bridge multiplier on both sides lies in* $(0,1)$*. The resulting contractions form a common full-dimensional regular global germ.*

Proof. For an isomorphic pair, transport one physical parameter section through the labelled isomorphism. For a T pair, use the physical analytic sections constructed in lemma 6.5, including when the unchanged context reconnects two triangle terminals inside one theta factor. Relative to the fixed analytic bridge slices, write the resulting positive incidence scales as $a^{(k)}_{v,e}$, where $k = 1, 2$ indexes the two networks. Shrink every local germ to a common compactly contained neighborhood. The finitely many scales and their reciprocals are then bounded.

For each bridge $e = uv$, choose a common effective scale z_e in a sufficiently small positive interval and define

$$x^{(k)}_e = \frac{z_e}{a^{(k)}_{u,e} a^{(k)}_{v,e}}.$$

Positivity gives $x^{(k)}_e > 0$, and the uniform lower bounds on the products $a^{(k)}_{u,e} a^{(k)}_{v,e}$ permit the same interval to make both multipliers less than one. Bridge choices are made edge by edge; the tree structure additionally ensures that the associated incidence normalizations have no holonomy. Incidence factors cancel in (4), so the two contractions agree. Finally, lemma 5.3 supplies an analytic extraction map inverse to contraction on this physical locus. Hence the product of the local germs and bridge intervals has the expected rank, and its common image is a full-dimensional regular global germ. □

7 Proof of the global theorem

Proof of theorem 1.1. Assume $N \preceq_{\mathrm{JC}} N'$. By corollary 4.5, their labelled decorated bridge trees agree. The exact bridge fiber and proposition 5.5 induce a source-relative projective containment for every pair of corresponding nontrivial factors. Root factors reduce to independently chosen real incoming ports by lemma 3.1. The local theorem then forces each pair to be labelled-isomorphic or T-related.

Conversely, lemma 6.6 applies to the corresponding local isomorphism and T germs. It produces one common full-dimensional regular global germ. This proves both the converse containment and $\bowtie_{\mathrm{JC}}$ statement. Since the right-hand condition is symmetric, no proper one-sided containment occurs. □

Proof of corollary 1.2. For fixed n there are finitely many topologies in the class. Tree-child paths give $r \le n-1$ reticulations, binary degree counting gives $t = n + r - 2$, and a rooted presentation has at most $4n - 3$ vertices.

For a fixed topology N, put $d_N = \dim \mathcal{V}^{\mathrm{JC}}_N$. Its complex model closure is irreducible as the closure of a polynomial image of an irreducible parameter space. For each competing topology N', define

$$Z_N(N') = \overline{\mathcal{M}^{\mathrm{JC}}_N \cap \mathcal{M}^{\mathrm{JC}}_{N'}}^{\,Z,\mathcal{V}_N},$$

the Zariski closure inside $\mathcal{V}_N^{\mathrm{JC}}$. Then let

$$E_{\mathrm{top}}(N) = \bigcup_{N' \not\sim_{\mathsf{T}} N} Z_N(N').$$

Every member is proper. Indeed, by lemma 5.4 and the cited semialgebraic dimension results, a semialgebraic subset of a nonempty smooth real d_N-stratum has dimension d_N exactly when it has nonempty relative interior there. Thus a nonproper $Z_N(N')$ would force the real semialgebraic intersection to have full dimension on a regular source stratum and hence relative interior, giving $N \preceq_{\mathrm{JC}} N'$ contrary to theorem 1.1. Tarski–Seidenberg makes every projected intersection semialgebraic [2, Theorem 2.2.1, applied iteratively]. Bochnak–Coste–Roy [2, Propositions 2.8.2, 2.8.5(i), and 2.8.13] imply that finite unions preserve the maximum dimension, that a lower-dimensional semialgebraic subset has proper Zariski closure, and that

$$\dim(\overline{A} \setminus A) < \dim A.$$

Enlarge this finite union by the singular locus and by the Zariski closures of images of parameter loci on which the Jacobian rank is less than d_N. Stratify each such locus into finitely many smooth semialgebraic pieces of constant rank. The constant-rank theorem bounds every image piece by that rank, and Tarski–Seidenberg preserves semialgebraicity; hence each image has dimension at most $d_N - 1$ and proper Zariski closure; semialgebraic bijections preserve dimension by Bochnak–Coste–Roy [2, Theorem 2.8.8]. Finally adjoin only the zero sets of the observable Fourier cut anchors and atlas-witness pullbacks that the exact crosswalk certifies to be nonidentically zero on N's intended regular stratum. This is a finite indexed list, not the union of every atlas polynomial. The resulting E_N is still proper. No unspecified physical-parameter factor is projected to distribution space. Outside E_N, every realizing topology is T-equivalent to N. □

8 Structural reconstruction

The proof gives an exact terminating reconstruction algorithm for a generic distribution known to arise in the class.

R1. Fourier-transform the pattern probabilities.

R2. Test split flattenings using theorem 4.3; recover the compatible cut system and build the labelled reduced bridge tree.

R3. Use (6) to distinguish ordinary degree-three components from three-sunlets.

R4. Factor the positive rank-one cut blocks and impose the analytic incidence normalizers from theorem 5.1; recover every projective local tensor orbit and normalized effective bridge scale.

R5. Evaluate the bounded graph-derived invariant deck and restoration certificates to identify one rigid support of each nontrivial factor modulo T.

R6. Use common-anchor one- and two-port probes to recover every segment and every total port order.

R7. Among the labelled T-orientations that are admissible members of S_{TC}, choose the lexicographically least one and assemble the canonical mixed graph.

Every loop is finite and every local probe uses at most ten tensor ports. A direct implementation uses at most $2^{n-1} - 1$ split tests and $O(n^{10})$ bounded local probes, excluding the bit complexity of exact algebraic-number operations. Thus we make no unproved polynomial-in-n claim, while the combinatorial operation count is polynomial in the length of the explicit 4^n input table. The algorithm returns the canonical topology modulo T; it does not recover physical bridge multipliers.

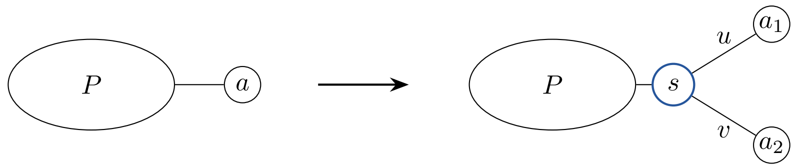


Figure 5: Identical cherry substitution at corresponding leaves. In Fourier coordinates the two new arms multiply the old port tensor, and positive ratios recover u, v, and the original tensor analytically. Each added leaf therefore contributes exactly two dimensions while leaving every nontrivial blob unchanged.

To list which orientations realize one particular distribution, one must test open-model membership for the finite T variants. The common-germ theorem does not imply complete stochastic-image equality.

9 Sharpness outside the strong class

We now prove theorem 1.3. The argument is included to make the boundary theorem self-contained; its exact verifier is also released as a frozen separately implemented package.

9.1 Identical leaf substitution

Replace a leaf a in both networks by the same cherry with new pendant multipliers $u, v \in (0,1)$. If P is the old tensor, then

$$\widetilde{P}(g_X, g_1, g_2) = P(g_X, g_1 \oplus g_2) u^{\mathbf{1}[g_1 \neq 0]} v^{\mathbf{1}[g_2 \neq 0]}. \tag{7}$$

For nonzero h,

$$uv = \widetilde{P}(0, h, h), \qquad \frac{u}{v} = \frac{\widetilde{P}(g_X, h, 0)}{\widetilde{P}(g_X, 0, h)},$$

where g_X has total h. Strict positivity gives a unique positive analytic inverse for u, v and then for every coordinate of P. Cherry substitution therefore adds exactly two dimensions and preserves a common regular open region.

9.2 The triangle-free Omega family

Let S be the root; let U, P_0, P_1, P_4 be tree vertices; and let V, X be reticulations. Both rooted presentations have arcs, in edge-parameter order,

$$\begin{gathered} U \to V,\ U \to P_0,\ P_0 \to P_1,\ P_1 \to V,\ S \to U,\ S \to X, \\ V \to P_4,\ P_4 \to X,\ P_0 \to L_0,\ P_1 \to L_1,\ P_4 \to L_2,\ X \to L_3. \end{gathered} \tag{8}$$

For Ω_4 label (L_0, L_1, L_2, L_3) by $(1,2,3,4)$; for Ω_4' use $(2,1,4,3)$.

Proposition 9.1 (Topology of the Omega pair)**.** *The fixed* sd_0 *reductions are simple, binary, triangle-free level-2 networks in* $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$*. They are labelled nonisomorphic and hence not* T*-equivalent.*

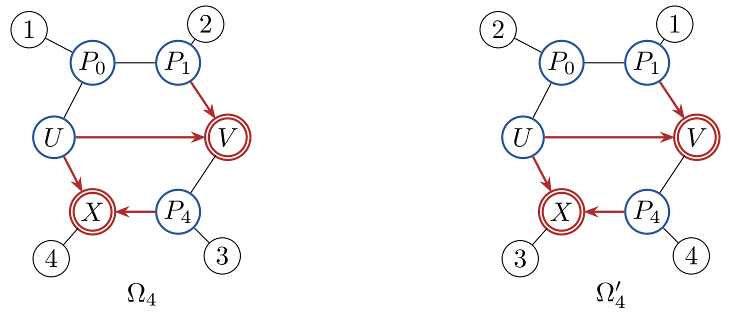


Figure 6: The triangle-free Omega sharpness pair. Both mixed graphs have cycle lengths $4, 4, 6$ and the same arrowhead pattern; the labelled pendant attachments are $(1, 2, 3, 4)$ on the left and $(2, 1, 4, 3)$ on the right. Each graph has seven admissible rootings, exactly two of which are tree-child. Up to reflection, both have quarnet type (2c) in the taxonomy of Englander et al. [6, Figure 5].

Proof. The displayed rootings are binary, LSA-valid, acyclic, stack-free, and tree-child. Their one-step reductions are already simple and have no parallel edges. Direct cycle enumeration in the one nontrivial blob gives lengths $4, 4, 6$, so the reductions are triangle-free and level two. Exhaustive orientation of the fixed mixed graph gives seven admissible rootings, exactly two tree-child. In each of the other five, U has both reticulations V, X as children. Equivalently, in the mixed graph U tails two retained edges and has only one undirected incidence; lemma 2.2 therefore gives weak but not strong tree-childness. In Ω_4, leaf 1 has unordered graph distances $\{3, 3\}$ to the two reticulations; in Ω'_4 they are $\{2, 4\}$. This label-preserving invariant proves nonisomorphism. There is no triangle on which T could act. $\square$

Proposition 9.2 (Exact Omega overlap)**.** *The open JC images of Ω_4 and Ω'_4 have dimension nine and share a regular relatively open nine-dimensional germ.*

Proof. The exact correspondence, strict rational point, and parameter ordering are recorded in appendix B. A clean-room displayed-tree calculation checks all 64 zero-sum Fourier identities symbolically under the nine-variable rational map and all 256 Fourier and inverse-pattern coordinates at the strict point. The fourteen nonconstant JC orbit representatives, together with the constant coordinate and JC state symmetries, determine all zero-sum coordinates. Every transition multiplier and inheritance probability lies in $(0, 1)$.

For the upper rank bound, set the four pendant multipliers to one and use the ten core coordinates $(a_0, \dots, a_7, \lambda_V, \lambda_X)$ in the edge order of (8). The Jacobian of the fourteen nonconstant orbit coordinates is 14×10. Exact row reduction over the rational-function field gives rank six. As a readable lower witness for that core calculation, at the strict source core coordinates in the pendant-one gauge, the orbit rows (A, B, C, D, E, F) and columns $(a_0, a_1, a_2, a_3, a_4, a_7)$ have determinant

$$-\frac{723}{8589934592} \neq 0.$$

The two displayed-rooting presentations have the same arc set and differ only by a permutation of the taxon labels. That leaf permutation induces an invertible permutation of the Fourier coordinates and hence a linear isomorphism between their model images. Their dimensions are therefore equal, so the source upper bound also holds for the target. Restoring the four pendant parameters adds at most four torus directions. The coordinatewise Euler identity

$$a_4 \frac{\partial c_g}{\partial a_4} + a_7 \frac{\partial c_g}{\partial a_7} = \mathbf{1}[g_4 \neq 0] c_g,$$

shows that the pendant direction for leaf 4 is already a linear combination of two core directions. Thus the complete model rank is at most $6 + 4 - 1 = 9$. The displayed-rooting source and target minors listed with their row and column sets in Supplement §5 give the reverse rank inequality and regularity. The corresponding alternative-rooting minors give an additional exact check that the rank certificate is invariant under the admissible rooted presentation. The rational correspondence is defined and remains strict on a neighborhood of the certified source point, and its common coordinate map has rank nine. Its image is therefore a common nine-dimensional local submanifold, relatively open in each nine-dimensional model image.

This calculation is also an explicit instance of the lower-than-naive model dimensions noted by Englander et al. [6, Discussion]. Here the pendant-one core has exact rank six, and the Euler relation exhibits one further dependency between a pendant torus direction and the core tangent space. □

The switching calculation gives a useful, but insufficient by itself, explanation of the collision. Both networks have the same labelled quartet-topology multiset: three switchings display $12 \mid 34$ and one displays $14 \mid 23$. At the certified gauge the four inheritance weights are equal, while the rational map in appendix B redistributes core and pendant edge factors. The equality is therefore not root relocation and still requires the complete Fourier and rank certificates.

Relation to Englander et al. Up to reflection, both Ω_4 and Ω_4' are differently labelled quarnets of type (2c) in Englander et al. [6, Figure 5]. Their Theorem 2.11 does not separate the pair because the two networks have the same displayed-quartet set (in fact, the same multiplicity pattern). Their Lemma 2.14 treats specified relabellings of type (1b), all distinct type (2a) quarnets, and distinct quarnets of types (3a), (3c), and (3d), but contains no type-(2c)-versus-type-(2c) distinction. Corollary 2.17 separates type (2c) from quarnets having a prune leaf, not from another type (2c) labelling. Finally, both Omega networks lie outside their strongly tree-child class: U tails two retained reticulation edges but has only one undirected incidence. Thus the collision is consistent with every applicable result there, while showing that the type-(2a) statement of Lemma 2.14(b) does not extend to all type-(2c) relabellings; the present obstruction lies outside their strongly tree-child class. The pair is also not galled, because V is ancestral to X inside their common blob, and its mixed topology is already simple under definition 2.1.

Proof of theorem 1.3. The four-leaf case is propositions 9.1 and 9.2. Repeatedly apply identical cherry substitution to one fixed leaf other than leaf 1, retaining that old label on one cherry child and assigning the next new taxon label to the other. Pendant substitution adds no blob or cycle, a tree-child rooting extends, and the original omnian rooting remains admissible. Contracting the common labelled pendant cluster would turn any isomorphism of the enlarged pair into an isomorphism of the four-leaf pair, so nonisomorphism persists. The common dimension is $9+2(n-4) = 2n+1$. □

9.3 A second mechanism: the triangle-containing Theta family

Theorem 9.3 (Theta pendant-transfer sharpness)**.** *For every $n \geq 4$ there are two nonisomorphic, non-T-equivalent networks $K_n, K_n' \in W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$ whose open JC images share a regular relatively open subset of their common full dimension $2n$.*

We construct the four-leaf base as follows. Let the internal vertices be a root ρ, tree vertices

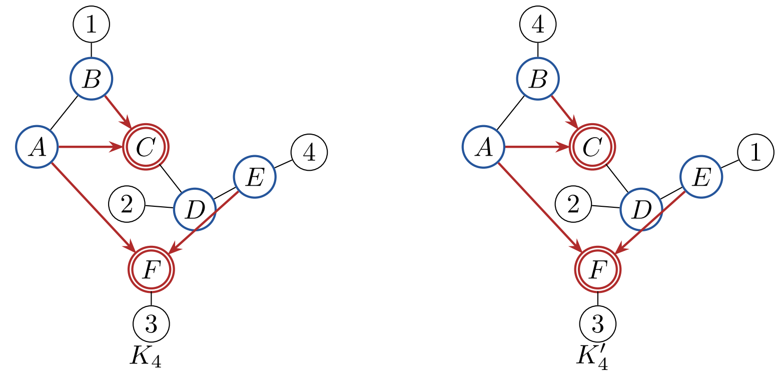


Figure 7: The triangle-containing Theta sharpness pair. The labelled attachments at B and E are exchanged. Leaf 1 is adjacent to a triangle vertex on the left and to a nontriangle vertex on the right, so this ambiguity is not ordinary triangle redirection.

A, B, D, E, and reticulations C, F. Both rooted networks have arcs

$$\rho \to A, \quad \rho \to C, \quad A \to B, \quad B \to C, \quad C \to D, \quad D \to E, \quad A \to F, \quad E \to F. \tag{9}$$

Their pendant arcs are

$$\begin{aligned} K_4 : &\quad B \to 1,\ D \to 2,\ F \to 3,\ E \to 4, \\ K_4' : &\quad E \to 1,\ D \to 2,\ F \to 3,\ B \to 4. \end{aligned}$$

After root suppression, the nontrivial blob is the theta with A–C paths

$$A - C, \qquad A - B - C, \qquad A - F - E - D - C,$$

whose cycle lengths are $3, 5, 6$.

Proposition 9.4 (Combinatorics of the weak pair)**.** *The two reductions lie in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$, are level two, and are neither isomorphic nor T-equivalent.*

Proof. The displayed rooted networks are binary, acyclic, and tree-child, proving weak tree-childness. Rerooting on $B \to C$ makes A the parent of both reticulations C, F; equivalently its mixed-graph tail fails lemma 2.2, so that admissible rooting is not tree-child. The blob contains exactly the two reticulations. In K_4, leaf 1 is adjacent to the triangle vertex B; in K_4', it is adjacent to the nontriangle vertex E. This labelled graph property is unchanged by isomorphism or triangle redirection. □

9.4 Exact Fourier overlap

Use the fourteen nonconstant four-leaf JC orbit coordinates A through O in the order fixed in appendix A. Both displayed-tree parameterizations satisfy

$$\begin{aligned} J - K - M + N &= 0, \\ J - AH - BF + CE &= 0, \\ GL - EN &= 0, \\ L^2 - BEH &= 0, \\ BM - DL - B^2F + BCE &= 0, \\ BEO - BGH - CEL + DEH &= 0. \end{aligned} \tag{10}$$

On $BE \neq 0$, five equations reconstruct J, K, M, N, O from $A, \ldots, H, L$, and the remaining equation is $L^2 = BEH$. Hence the common localized ring is an irreducible eight-dimensional domain and is smooth on the physical locus because $\partial(L^2 - BEH)/\partial H = -BE \neq 0$.

At one strict source point, the common first eight and final six coordinates are

$$(A, \ldots, H) = \left(\frac{277}{8000}, \frac{3}{40}, \frac{67}{2400}, \frac{127}{16000}, \frac{27}{5000}, \frac{81}{5000}, \frac{153}{160000}, \frac{9}{500}\right),$$

$$(J, \ldots, O) = \left(\frac{27}{16000}, \frac{261}{320000}, \frac{27}{10000}, \frac{27}{20000}, \frac{153}{320000}, \frac{183}{80000}\right).$$

The target parameters lie in $\mathbb{Q}(\beta)$, where β is the smaller root of

$$43337075\beta^2 - 36083110\beta + 7336259 = 0, \qquad \frac{441}{1250} < \beta < \frac{3529}{10000}. \tag{11}$$

An exact isolating-interval calculation puts every edge multiplier and both inheritance probabilities strictly in $(0, 1)$. The complete target point and its elementary interval proof are printed in Section 6 of the supplement; the frozen verifier checks all 256 Fourier coordinates in $\mathbb{Q}(\beta)$.

For K_4, an eight-coordinate gauge Jacobian determinant factors as

$$-\frac{P^3 s^3 Q^4 t R^4 u^3 v S^3 (s-1)^2 (v-1)(v+1)^2}{16384},$$

and for K_4' the corresponding determinant is

$$-\frac{(P')^3 x^2 y^2 z (R')^4 w^4 (S')^3 (Q')^4 (x-1)^2 (z-1)(z+1)^3}{32768}.$$

Both are nonzero at the common point. Thus both images are eight-dimensional local submanifolds of the same smooth irreducible locus. Since projection to $(A, \ldots, H)$ is a local coordinate system on the positive smooth branch $L = \sqrt{BEH}$, rank eight of this projection for each physical parameterization implies that each image contains a relative neighborhood of the common point in that same branch. The reconstruction equations then force the remaining coordinates from the first eight.

Proof of theorem 9.3. The four-leaf case is proposition 9.4 and the exact overlap above. Repeatedly substitute a cherry in the same corresponding leaf other than leaf 1 in both networks. The model dimension becomes $8+2(n-4) = 2n$. The displayed tree-child rootings extend, the original omnian witness remains, and leaf 1 stays adjacent to a triangle vertex on one side and a nontriangle vertex on the other. Thus the pair remains in $W_{\mathrm{TC}} \setminus S_{\mathrm{TC}}$, nonisomorphic, and non-T-equivalent for every $n \geq 4$. $\square$

10 Biological consequence and scope

Within the standard strongly tree-child level-2 class, generic exact infinite-data JC observations recover the bridge tree, every labelled descendant attachment, every cycle/theta core, and every port order. The only generic topological ambiguity is which vertex of an ordinary triangle carries the reticulation arrowheads. Physical bridge multipliers are not recovered by the topology theorem, and compatible triangle orientations need not realize the same distribution outside their common germ.

Strong tree-childness is essential. Immediately in the larger weakly tree-child class, the Omega family has a full-dimensional regular JC ambiguity despite being triangle-free. The Theta family separately moves a labelled component between a triangle vertex and a nontriangle vertex. Thus the boundary is not merely a change in root-presentation multiplicity and is not caused only by the ordinary triangle quotient.

What the data recover. Away from the stated proper algebraic exceptional set, the exact distribution recovers every cut split, the labelled reduced bridge tree, cycle versus theta blob type, every labelled descendant attachment, every port order, and the standard semi-directed topology modulo ordinary triangle redirection.

What the theorem does not recover. It does not identify physical bridge multipliers or all numerical parameters; it does not assert one triangle orientation compatible with every distribution, classify the exceptional locus pointwise, address richer substitution models, or supply a finite-sample statistical procedure.

11 Reproducibility and proof status

The release contains three deterministic entry points: a quick theorem crosswalk, a full replay of all committed exact certificates, and a bounded regeneration from primitive graph encodings. The primary and clean-room implementations share no graph canonicalizer, displayed-switching engine, descendant-mask code, separator selector, or rank routine at the load-bearing gates. Mutation suites reject relation deletion, changed port matching, source–target reversal, polynomial reassignment, broken restoration/probe parentage, reciprocal-only bridge gauges, and false physical-parameter recovery. The separate Omega suite also rejects an incomplete rooting list, testing only the displayed rooting, changed arrowheads or parents, boundary parameters, altered Fourier entries or rank minors, and a triangle-free claim without cycle enumeration.

Historical failed certificates are preserved but excluded from every active manifest. These include the reciprocal-only bridge chart, schema keys that merged distinct rooted relations, ineffective mutations, and attempts to promote weak-class Theta or Omega phenomena into the locked strong class. The whole-proof release checker is a fail-closed consistency gate; the mathematical evidence comes from the graph-derived component certificates and their separately implemented replays. The exact theorem-to-certificate crosswalk and all file hashes accompany the release.

A Four-leaf JC orbit coordinates

Represent the nonzero elements of $\mathcal{G}$ by $1, 2, 3$. The fourteen nonconstant JC orbit representatives are

symbol	tuple	name	symbol	tuple	name
A	0011	r_{34}	B	0101	r_{24}
C	0110	r_{23}	D	0123	t_{234}
E	1001	r_{14}	F	1010	r_{13}
G	1023	t_{134}	H	1100	r_{12}
J	1111	u_{1111}	K	1122	u_{1122}
L	1203	t_{124}	M	1212	u_{1212}
N	1221	u_{1221}	O	1230	t_{123}

The seven bounded-atlas invariant templates are stored as sparse lists

$$\sum_m c_m \prod_{j\in m} q_j,$$

with coordinate indices in this orbit convention. Their complete coefficient lists are released in `jc_root_spanning_atlas_data.py` and `seventh_invariant.json`. The theorem verifier first transports these inert coefficient lists through every ordered leaf map and then regenerates each pullback from the bound graph tensor. No result is accepted by looking up a topology identifier.

B Exact Omega correspondence

Index the twelve edges by the order in (8). On the source set

$$a_5 = a_6 = a_{11} = \lambda_V = \lambda_X = \frac{1}{2},$$

write

$$(u_0, u_1, u_2, u_3, u_4, u_5, p_0, p_1, p_2) = (a_0, a_1, a_2, a_3, a_4, a_7, a_8, a_9, a_{10}), \quad g = u_4 + 2u_5, \quad h = u_0u_5 + 2u_4.$$

The target correspondence is

$$\begin{aligned}
b_0 &= \frac{2u_5(4-u_0)}{g}, & b_1 &= \frac{u_3}{4-u_0}, & b_2 &= u_2, & b_3 &= \frac{2u_1h}{g},\\
b_4 &= \frac{4u_4p_2(4-u_0)}{h}, & b_5 &= \frac{1}{2}, & b_6 &= \frac{1}{2}, & b_7 &= \frac{2u_0p_2g}{h},\\
b_8 &= p_1, & b_9 &= p_0, & b_{10} &= \frac{g}{8}, & b_{11} &= \frac{1}{2}, \qquad \mu_V = \mu_X = \frac{1}{2}.
\end{aligned}$$

Direct substitution into the independently generated displayed-tree tensors annihilates all 64 zero-sum coordinate differences.

The strict source and target points are, respectively,

$$\left(\frac{1}{2}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{20}, \frac{1}{2}, \frac{1}{2}, \frac{1}{10}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right)$$

and

$$\left(\frac{7}{12}, \frac{1}{7}, \frac{1}{2}, \frac{41}{48}, \frac{28}{41}, \frac{1}{2}, \frac{1}{2}, \frac{12}{205}, \frac{1}{2}, \frac{1}{2}, \frac{3}{40}, \frac{1}{2}; \frac{1}{2}, \frac{1}{2}\right).$$

At the source point the three denominators are $g = 3/5$, $h = 41/40$, and $4 - u_0 = 7/2$. The independent rank-nine minors are

$$-\frac{171}{23058430092136939520000000}, \qquad -\frac{513}{92233720368547758080000000}$$

for the displayed source and target. The corresponding alternative-rooting minors are

$$\frac{57}{5764607523034234880000000}, \qquad \frac{189}{23058430092136939520000000}.$$

Inverse Fourier transformation gives the same 256 strictly positive pattern probabilities on both sides; the smallest is 87763/26214400.

C Certificate crosswalk

All paths in this appendix are relative to the extracted proof-certificate bundle archived in [13]. The bundle's top-level `README_FIRST.md` gives the single entry point and its `ACTIVE_MANIFEST.json` authenticates every listed file.

Claim	Minimal exact evidence
Convention, primitive supports, cut recovery, and bridge fibre	`primary/certificates/core_universe.json`, `primary/certificates/support_universe.json`, `reviews/root_probe/`, and `independent/bridge_cut/`.
Theorem 6.3, including restoration and coherent probes	`atlas/ATLAS_EVIDENCE_BINDINGS.jsonl.gz`, whose theorem rows point to `atlas/RESTORATION_CLOSURE_BINDINGS.jsonl.gz` or `atlas/DIRECT_ANCHOR_CLOSURE_BINDINGS.jsonl.gz`; restoration terminals point onward to `atlas/COMPACT_PATH_CLOSURE_BINDINGS.jsonl.gz`, binding every exact path, transport, witness, and polynomial record.
Complete deterministic regeneration from primitive inputs	`verifiers/regenerate_load_bearing.py`, invoked twice by `verify.sh` in `regenerate-all` mode.
Ordinary triangle common germ	`primary/certificates/jc_triangle_redirection_active.json` and `reviews/triangle_redirection_cleanroom/`.
Omega and Theta sharpness families	`sharpness/omega/` and `sharpness/theta/`.
Bundle integrity, totality, deterministic regeneration, and mutation tests	`ACTIVE_MANIFEST.json`, all four `atlas/*BINDINGS.jsonl.gz` streams, `expected_outputs/`, and `verify.sh` in all three modes.

Data and code availability

The load-bearing finite atlas, exact per-relation certificates, restoration and probe records, primitive graph inputs, primary verifier, and separately implemented replay verifier are archived in the curated proof-certificate bundle [13]. The small verifier capsule supplied with the preprint contains entry points, environment information, and checksums only; it is not the complete proof archive. No specialized phylogenetic-network classification package is required.

Author contributions

Alec Kriebel conceived and directed the study, fixed the mathematical scope and conventions, adjudicated proof and audit results, curated the reproducibility release, revised the manuscript and figures, and accepts responsibility for the final claims and submission.

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Competing interests

The author declares no competing interests.

Use of generative AI

Generative AI systems assisted with mathematical discovery and exploratory algebra, code generation and implementation, drafting, and adversarial review. Several earlier automated claims failed adversarial audit, including a reciprocal-only bridge chart and candidate ambiguity moves outside the locked standard strong class; those claims are withdrawn and preserved only in quarantined history. Here “separately implemented replay” means a code-independent implementation, not an independent human review. The active theorem is supported by exact graph-derived certificates and such separately written replays. No human specialist review is claimed. The author accepts responsibility for the mathematical statement, code, and final submission.

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